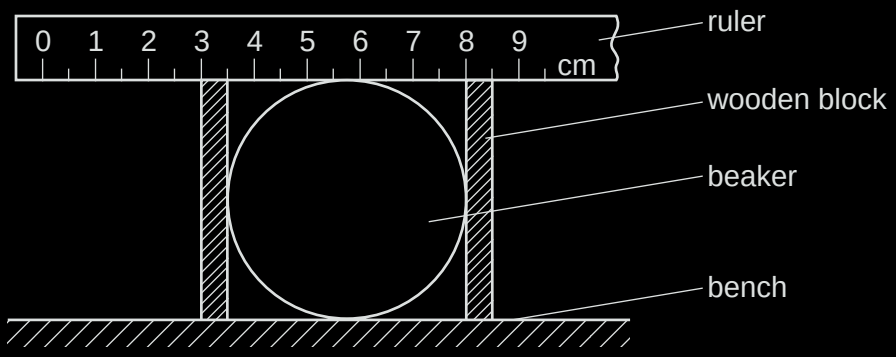


What is the diameter of the beaker?

A 4.5 cm **B** 5.0 cm **C** 5.5 cm **D** 8.0 cm

2 A student studies some equations.

power = work / time

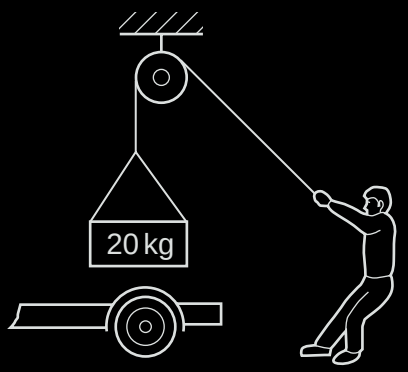
force = mass × acceleration

velocity = displacement / time

How many vector quantities are contained in the equations?

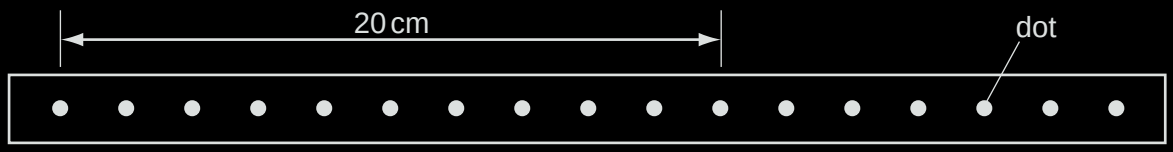
A 1 **B** 2 **C** 3 **D** 4

3 A person just supports a mass of 20 kg suspended from a rope.



What is the resultant force acting on the mass?

A 0 N **B** 10 N **C** 20 N **D** 200 N



What was the speed of the object?

A 2.0 cm / s **B** 5.0 cm / s **C** 100 cm / s **D** 200 cm / s

6 Three objects are cut from the same sheet of steel. They are different shapes but they all have the same mass.



disc square L-shape

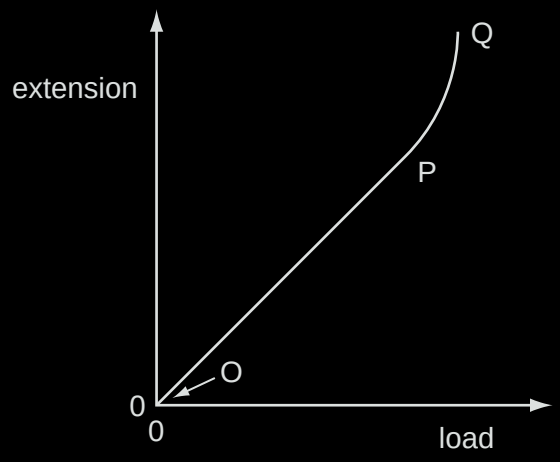
Which object has the greatest density?

A the disc

B the L-shape

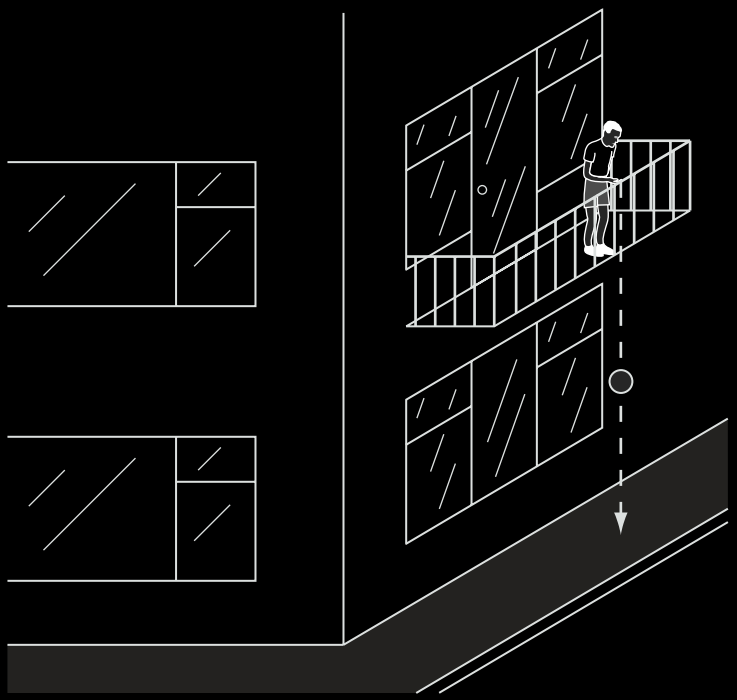
C the square

D they all have the same density



A O **B** OP **C** P **D** PQ

8 A young child holds a ball over the edge of a balcony. The ball has gravitational potential energy. The ball is then released. It falls onto a concrete path below, and bounces back up.



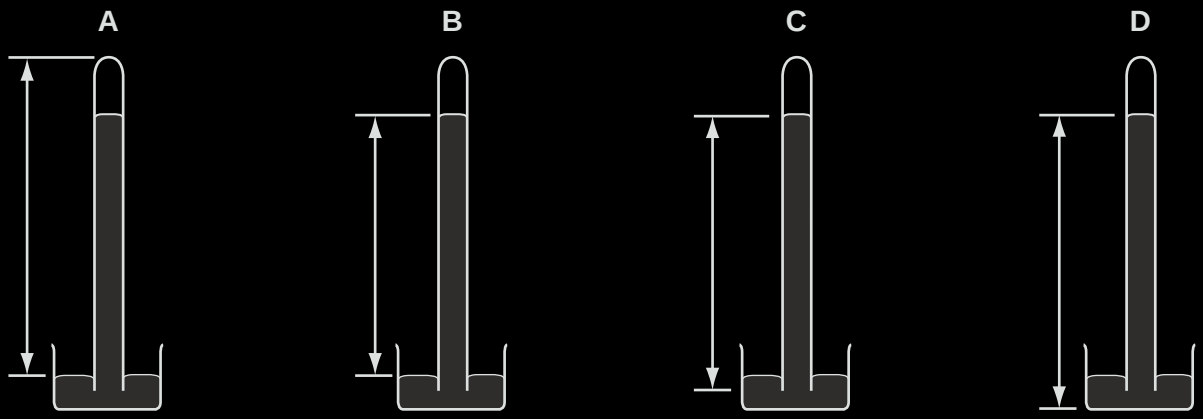
Which sequence represents, in the correct order, the transformations of the gravitational potential energy after the ball is released?

A → elastic potential energy → kinetic energy → chemical potential energy

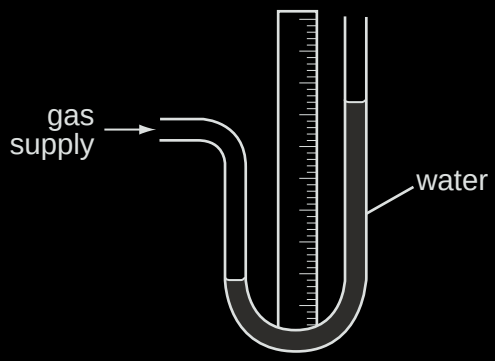
B → elastic potential energy → kinetic energy → gravitational potential energy

C → kinetic energy → elastic potential energy → kinetic energy

D → kinetic energy → gravitational potential energy → kinetic energy

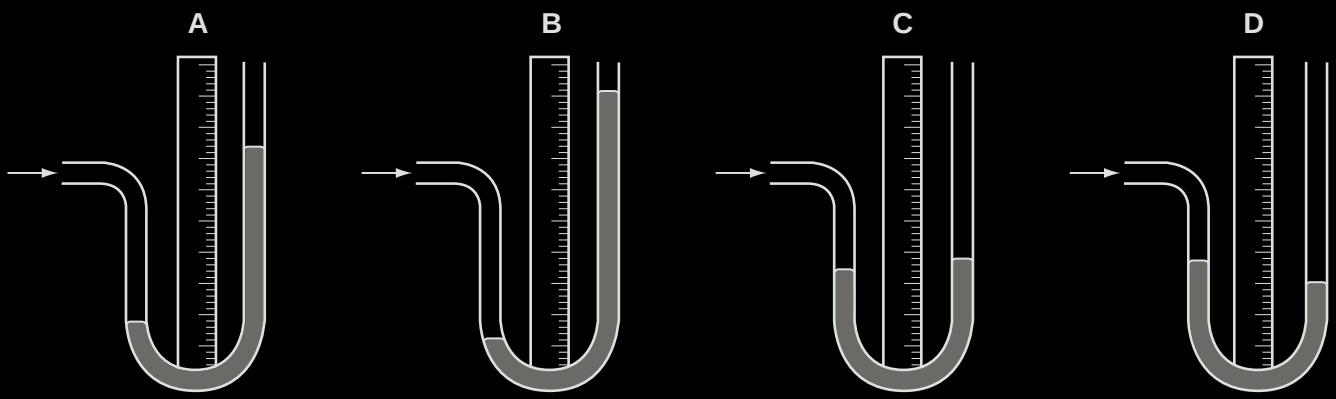


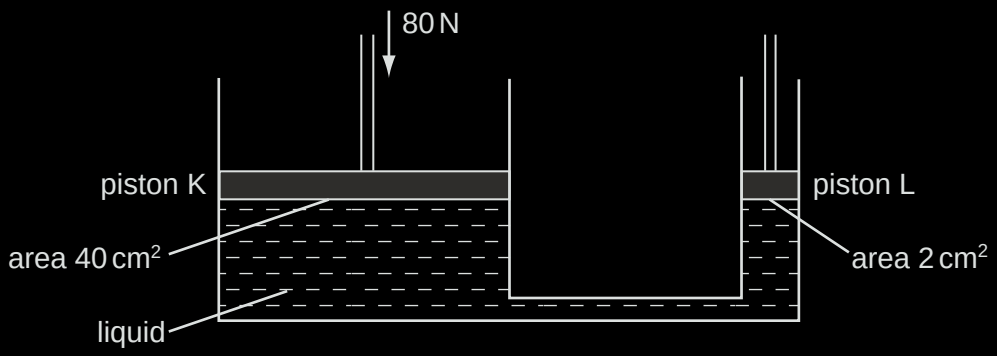
11 A water manometer is connected to a gas supply. The diagram shows the water levels.



The water is replaced by mercury, which is more dense than water.

Which diagram shows the mercury levels when the manometer is connected to the same gas supply?





A downward force of 80 N is exerted on piston K.

What will be the upward force exerted by the liquid on piston L?

A 1 N

B 4 N

C 80 N

D 1600 N

13 What is a property of both liquids and gases?

A They always fill their containers.

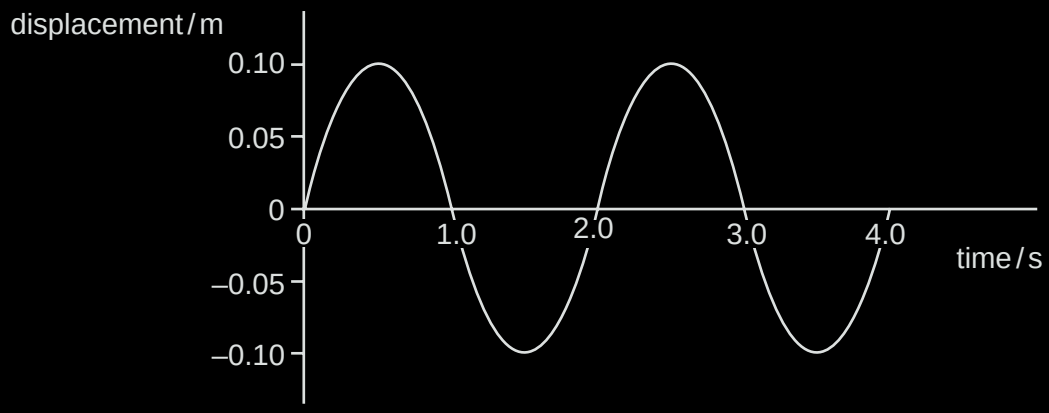
B They are incompressible.

C They can flow.

D They have molecules in fixed positions.

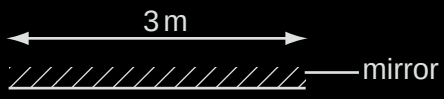
14 What conditions of temperature and surface area would produce the most rapid evaporation from a pool of water on a road surface?

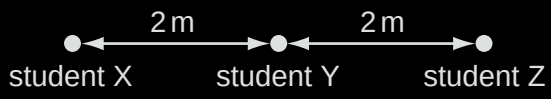
	temperature	surface area
A	high	large
B	high	small
C	low	large
D	low	small



What is the frequency of this wave?

A 0.25 Hz **B** 0.50 Hz **C** 1.0 Hz **D** 2.0 Hz





Student Y is standing opposite the mid-point of the mirror.

How many students can see the images of the other two?

A 0 **B** 1 **C** 2 **D** 3

22 The human eye has a converging lens system that produces an image at the back of the eye.

An eye views a distant object. What type of image is produced?

A real, erect, same size

B real, inverted, diminished

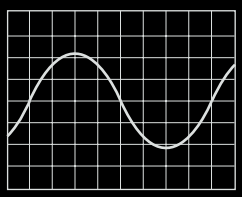
C virtual, erect, diminished

D virtual, inverted, magnified

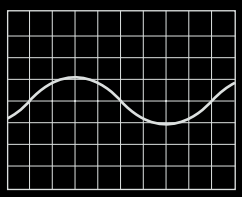
23 The diagrams show oscilloscope traces of sounds picked up by microphones. The oscilloscope controls are set in the same position for all the traces.

Which trace shows the sound that is both loud and low-pitched?

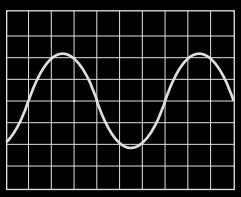
A



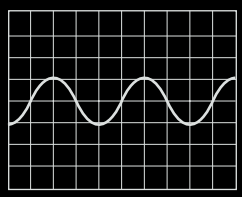
B

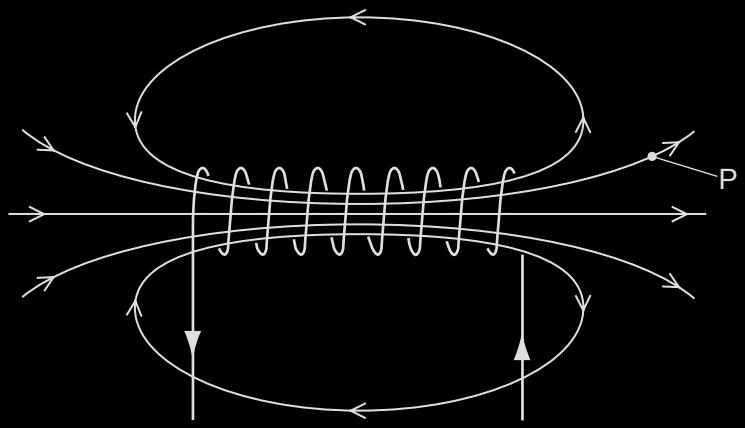


C



D

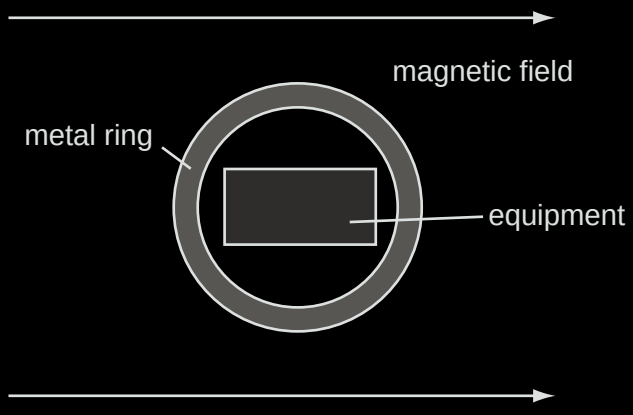




What is the effect on the magnetic field at the point P of using a larger current in the opposite direction?

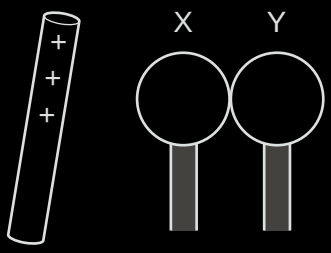
	field strength	field direction
A	decreases	reverses
B	decreases	unchanged
C	increases	reverses
D	increases	unchanged

25 A metal ring screens a piece of equipment from a magnetic field.



Which metal should be used for the ring, and why?

	metal	reason
A	copper	the metal carries the field lines around the equipment
B	copper	the metal is non-magnetic
C	iron	the metal carries the field lines around the equipment
D	iron	the metal is non-magnetic



What will be the charge on Y?

A negative and smaller than that on X

B negative and the same size as that on X

C positive and smaller than that on X

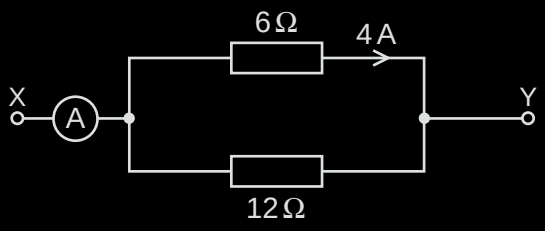
D positive and the same size as that on X

27 A wire has a resistance of 8 Ω. A second wire, made of the same material, has half the length and twice the cross-sectional area.

What is the resistance of the second wire?

A 1 Ω **B** 2 Ω **C** 8 Ω **D** 16 Ω

28 Two resistors of 6 Ω and 12 Ω are arranged in parallel. A p.d. is connected across the terminals X and Y. The current through the 6 Ω resistor is 4 A.



What is the current in the ammeter?

A 4 A **B** 6 A **C** 8 A **D** 12 A

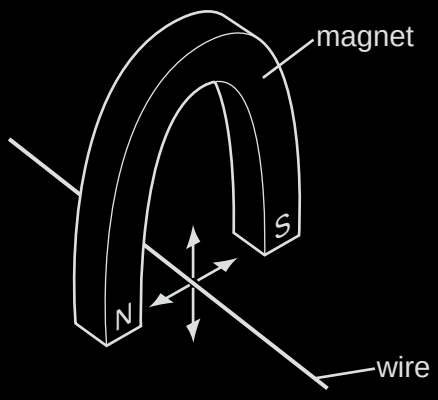
29 Which of the following would cost the **least** if operated from the same voltage supply?

A a 5000 W electric cooker used for 1 minute

B a 1000 W electric fire used for 10 minutes

C a 500 W electric iron used for 1 hour

D a 100 W lamp used for 1 day

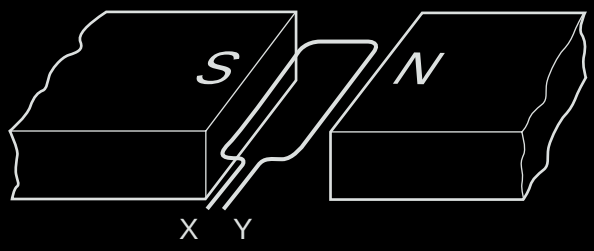


The current in the wire can be reversed. The poles of the magnet can also be changed over.

In how many of the four directions shown can the force act on the wire?

A 1 **B** 2 **C** 3 **D** 4

32 The diagram shows a coil in a magnetic field.



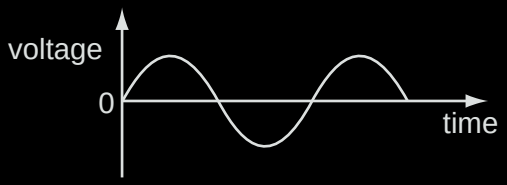
When the coil is part of a d.c. motor, what must be connected directly to X and Y?

A d.c. supply

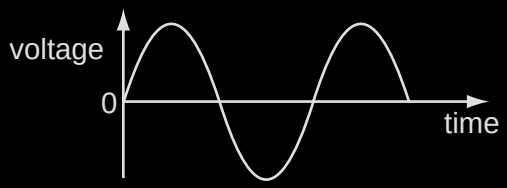
B slip rings

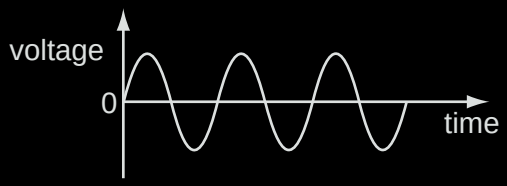
C soft-iron core

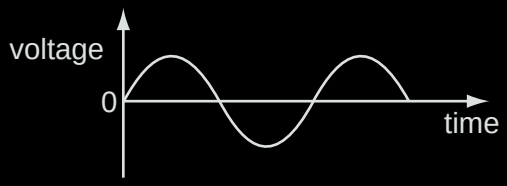
D split-ring commutator

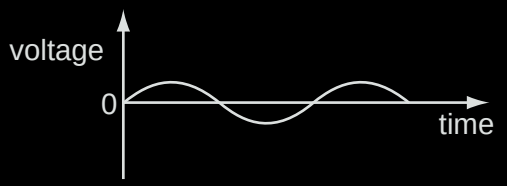


Which graph, drawn to the same scale as the input graph, shows how the output voltage varies with time?

A


B


C


D


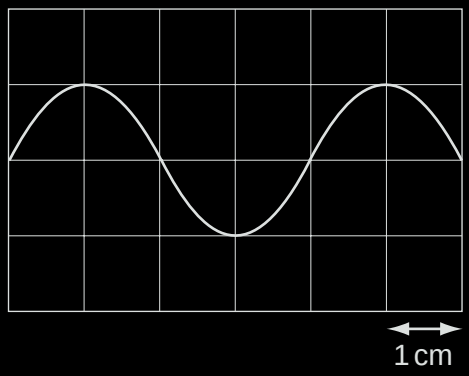
34 Why is electrical energy usually transmitted at high voltage?

A As little energy as possible is wasted in the transmission cables.

B The current in the transmission cables is as large as possible.

C The resistance of the transmission cables is as small as possible.

D The transmission system does not require transformers.



What is the time for one complete rotation of the coil of the generator?

A 5 ms **B** 10 ms **C** 20 ms **D** 30 ms

36 A capacitor can be made to store energy by connecting it to a d.c. power supply.



Along which paths do electrons flow as the capacitor starts to charge?

A from X to plate 1 only

B from X to plate 1 and from plate 2 to Y

C from Y to plate 2 and from plate 1 to X

D from Y to plate 2 only

37 A nucleus of sodium, Na, has 11 protons and 12 neutrons.

Which symbol represents this nucleus?

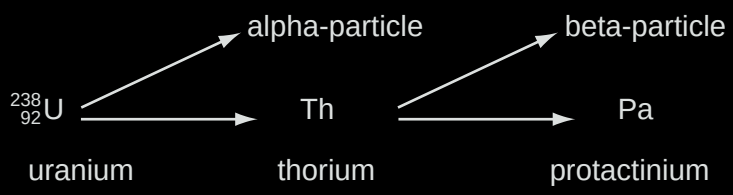
A $^{11}_{12}\mathrm{Na}$ **B** $^{12}_{11}\mathrm{Na}$ **C** $^{23}_{11}\mathrm{Na}$ **D** $^{23}_{12}\mathrm{Na}$

38 The half-life of a radioactive material is 24 years.

The activity of a sample falls to a fraction of its initial value after 72 years.

What is the fraction?

A 1/3 **B** 1/4 **C** 1/6 **D** 1/8



What is the proton number (atomic number) of protactinium?

A 95 **B** 91 **C** 90 **D** 89